

EEE412 Microwave Electronics-Lab Assignment 2

Preliminary Work

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"I completed this homework assignment independently."

Part 1

In this part we observe the 1 dB compression point analysis for the simplest amplifier design.

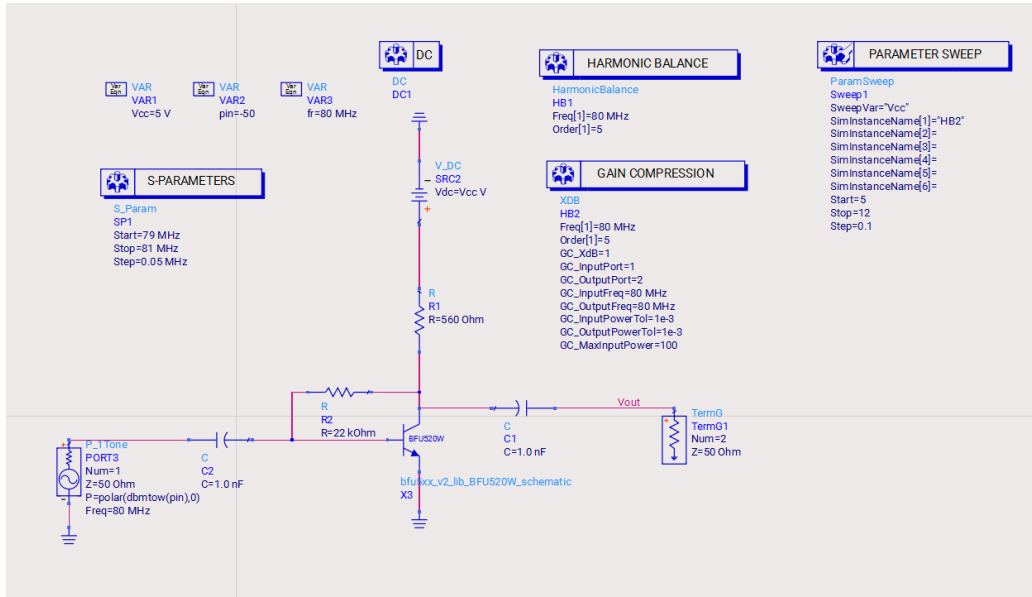


Figure 1: ADS schematic for P1dB analysis (when $f_r = 80$ MHz)

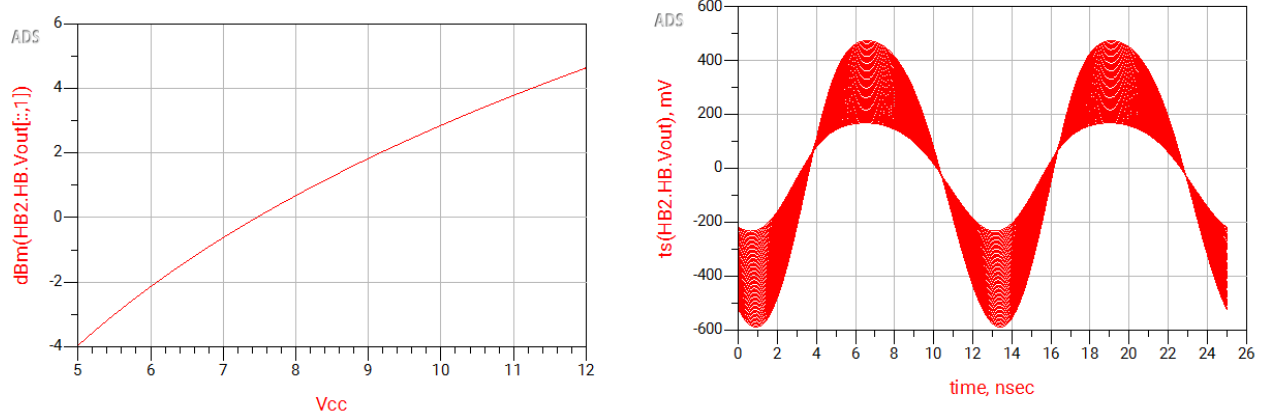


Figure 2: Time-domain Waveforms of the Output Voltage at the P1dB Point

The amplifier was analyzed using harmonic balance simulation at an input frequency of 80 MHz. The supply voltage V_{CC} was swept from 5 V to 12 V, while the input power was automatically controlled by the XDB block to determine the 1-dB gain compression point. The simulation results show that the extracted P_{1dB} increases as V_{CC} is raised.

Part 2

In this part we will observe the Output Third-Order Intercept Point (OIP3).

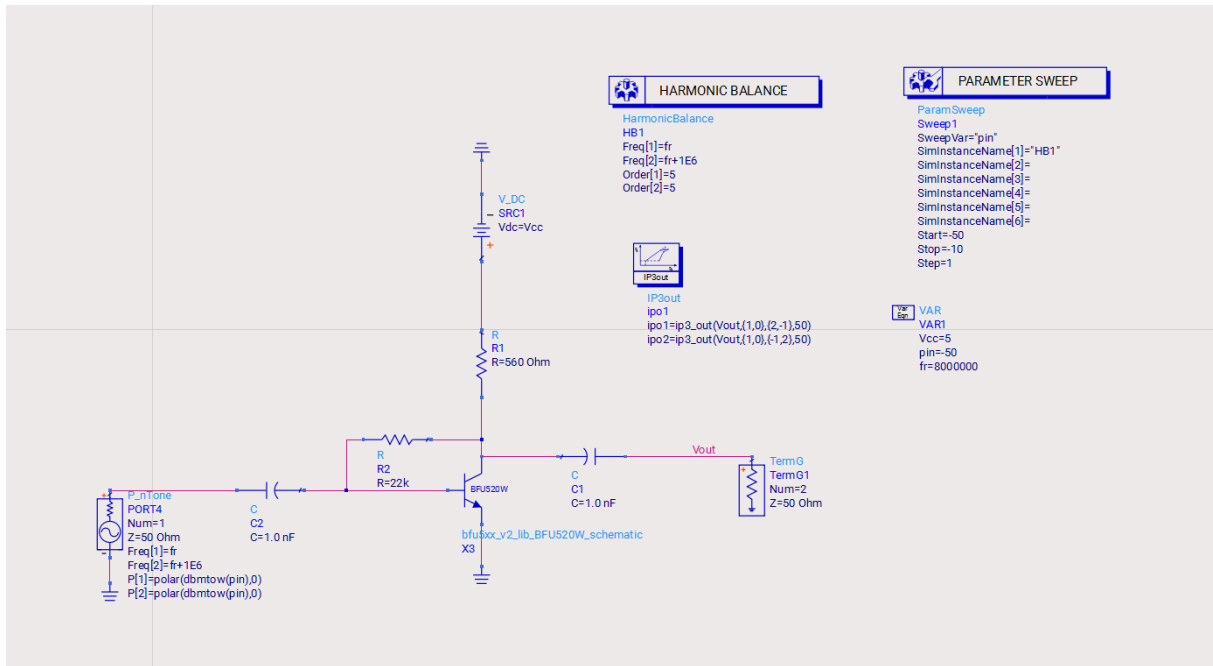


Figure 3: ADS schematic for OIP3 analysis (when $f_r = 80$ MHz)

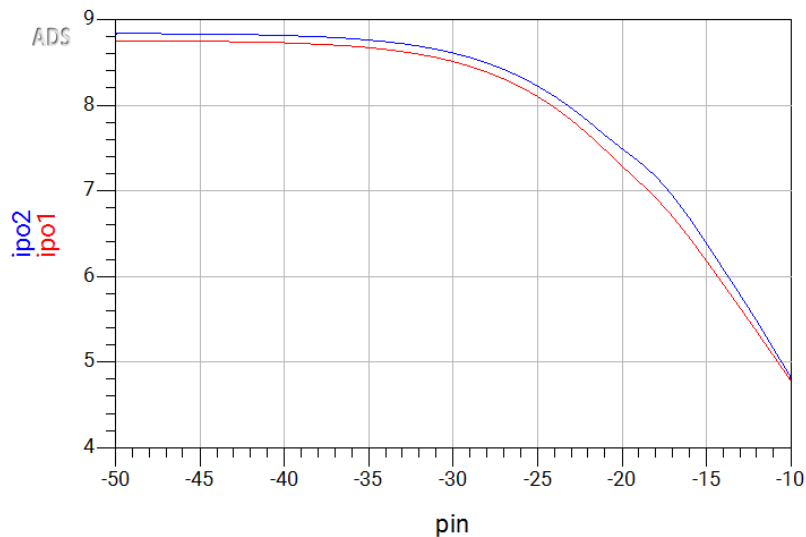


Figure 4: OIP3 Values with respect to Input Power

For the OIP3 analysis, the supply voltage was fixed at $V_{CC} = 5$ V. The amplifier was driven by a two-tone input with frequencies $f_1 = f_r$ and $f_2 = f_r + 1$ MHz, where $f_r = 80$ MHz. The input power was swept from -50 dBm to -10 dBm, and the output third-order intercept points were extracted using the IP3out block.